PRELIMINARY ANALYSIS:

Maritime Vessel Activity During The Ongoing Iran War

Persian Gulf and Gulf of Oman — March 2026

Prepared by: Jonathan Markey
Date: 11/04/2026

Executive Summary

This report presents a preliminary open-source investigation into maritime vessel activity across the Persian Gulf and Gulf of Oman during March 2026, using AIS vessel presence data, SAR vessel detections from the Global Fishing Watch API, and Sentinel-2 satellite imagery from the Copernicus Data Space.

AIS activity declined 55% between 1 and 13 March, coinciding with the escalation of hostilities in the Strait of Hormuz and the removal of vessel insurance for the area. Activity did not return to pre-drop levels, and 54% of identified vessels remained in the area for 15 or more days, consistent with reporting that daily Strait transits fell from 138 pre-conflict to single digits by late March.

Analysis of 1,198 dark vessel candidates identified a geographic shift from the western Gulf, where AIS activity concentrates, toward the eastern Gulf and the Strait of Hormuz. The Strait recorded the highest concentration of dark vessel candidates at 124, including 21 high-confidence detections, making it the priority area for further investigation.

Satellite imagery of Bandar Abbas Naval Base on 18 March revealed over 100 vessels visually present against 20 identifiable AIS records in the same time window, highlighting limitations in the available data resolution and the potential scale of undeclared vessel activity at Iranian ports.

The investigation is constrained by the low spatial resolution of the available data (0.1 degree grid of 11km), incomplete vessel identification in 69.3% of AIS records, and the use of open sources only. Findings are preliminary and all dark vessel identifications are candidates requiring further corroboration. Recommended next steps include acquiring higher-resolution AIS data, expanding satellite coverage to the Strait of Hormuz and other priority zones, and correlating vessel activity patterns with conflict event reporting.

Table of Contents

Table of Contents 2
Introduction 3
Methodology 3
1. AIS Vessel Presence 5
1.1 Spatial Distribution 5
1.2 Temporal Patterns 6
1.3 Vessel Persistence 7
2. Dark Vessel Candidates 8
1.1 Spatial Comparison 8
1.2 Geographic Concentration 9
3. Satellite Imagery — Bandar Abbas Naval Base 10
Findings and Caveats 12

Introduction

This report presents an investigation into maritime vessel activity across the Persian Gulf and the Gulf of Oman waters during March 2026, amid the ongoing Iran War. The intent of this report is not to produce a definitive intelligence assessment for the conflict but to demonstrate the application of multiple open data sources to characterise the maritime picture for the area.

The investigation examines three questions:

- What does the pattern of declared vessel traffic look like across the analysed area during this period and how has it changed over the course of the ongoing war?
- Where are vessels mainly operating in the area, and how does the clustering of vessels correlate with the strategic locations in the area?
- Can the data and satellite imagery be used in conjunction to supplement the vessel tracking for locations of interest?

Three data sources are used to address these questions. The first two sources are the Automatic Identification System (AIS) vessel presence data, and Synthetic Aperture Radar (SAR) vessel detection data, both of which were sourced from the Global Fishing Watch API (https://globalfishingwatch.org/). The third data source is Sentinel-2 satellite imagery accessed via the Copernicus data space. The area analysed covers the Persian Gulf and Gulf of Oman located in the bounded area of 47–63 degrees East and 22–30.5 degrees North.

The report is structured into three sections. Section 1 analyses the AIS vessel presence in the area to establish a baseline of declared maritime activity. Section 2 identifies and maps the dark vessel candidates, which are vessels defined as detected by radar but could not be matched to a broadcasted AIS signal. Section 3 presents satellite imagery of Bandar Abbas Naval Base, cross-referenced with AIS data from the same date and general time. Findings and caveats are summarised in the conclusion.

Methodology

Data Sources

Source	Type	Resolution	Period	Access
Global Fishing Watch	AIS Presence Report, Vessel ID	LOW, 0.1° grid, hourly	1–31 Mar 2026	GFW API
Global Fishing Watch	SAR Presence Report, MMSI	LOW, 0.1° grid, hourly	1–31 Mar 2026	GFW API
Copernicus	Sentinel-2 L2A TCI	10m optical	Selected passes	STAC API

AIS & SAR Datasets

The AIS presence dataset contains hourly vessel position reports across the analysed area for the target period. Each entry contains information for the timestamp, longitude, latitude, flag, gear type and the various vessel identifications such as MMSI, call sign and ship name. AIS is broadcasted from a ship via an onboard transponder, meaning that only vessels with active and functional transponders will appear in the dataset. Vessels not transmitting their AIS signal will not be identified. The dataset is retrieved at a low spatial resolution of 0.1 degrees, which aggregates the vessel positions into 11km wide grid cells. A significant proportion of the vessels in the dataset contain no vessel identification information and are classified as unidentified for the

purposes of this analysis. There are 1,865,149 rows of data in the AIS presence report for March 2026.

The SAR dataset is structured similarly to the AIS dataset, however unlike AIS the detections in this dataset rely on Sentinel-1 radar passes to detect physical objects on the water surface, identifying vessels regardless of whether they are broadcasting their position. Each SAR record includes a detection timestamp, position, MMSI where matched, and a detection count representing the number of times a vessel was detected at that location. SAR coverage is not continuous as it is dependent on satellite overpass times, meaning the dataset reflects periodic snapshots of the area rather than persistent surveillance. There are 8,878 rows of data in the SAR report for March 2026.

Dark Vessel Identification

SAR detects physical vessel presence regardless of AIS broadcast status. A join between SAR detections and AIS presence at the same 0.1° grid cell and date and time identifies probable SAR returns with no corresponding AIS signal. These are flagged as dark vessel candidates. Candidates with three or more SAR detections are classified as high-confidence to filter radar noise and small craft.

A limitation of the dark vessel identification methodology is that it relies on the vessel not being in the general location of AIS transmitting vessels at the same grid cell and time. This means that locations with high AIS transmission density such as ports can result in false negatives, where a dark vessel is not flagged because it shares a grid cell with an AIS-broadcasting vessel. Conversely, areas with sparse AIS coverage may produce false positives from radar returns off non-vessel objects. From the 8,878 SAR detections, 1,198 dark vessel candidates were identified.

Tools

Python (pandas, matplotlib, scikit-learn, rasterio, contextily, folium, plotly, datetime), Global Fishing Watch API via gfwapiclient Python package, Copernicus STAC API for Sentinel-2 satellite imagery via pystac_client.

1. AIS Vessel Presence

This section examines AIS broadcasting vessel traffic across the region during March 2026 to establish a baseline of declared maritime activity.

1.1 Spatial Distribution

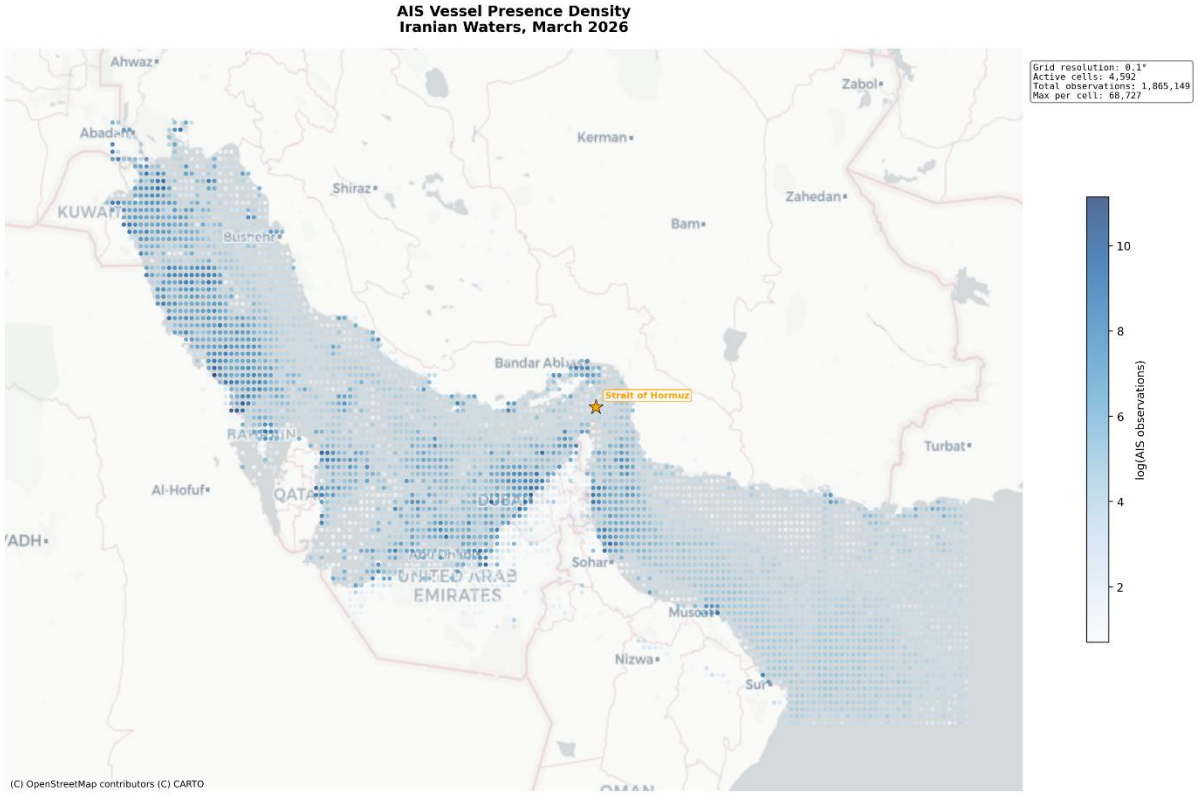


Figure 1.1: AIS Vessel Presence Density — Iranian Waters, March 2026

Figure 1.1 shows the spatial distribution of AIS vessel presence across the analysed area for March 2026. The density is displayed on a logarithmic scale across 4,592 active grid cells containing all the AIS data observations. The Western Persian Gulf area located to the North-West of Abu Dhabi accounts for 66.1% of all AIS observations. The densest single grid cell, of 68,727 observations, is also located here at 27.0 degrees North by 49.7 degrees East. Dubai and Jebel Ali represent the single largest activity hub at 6.1% of total observations followed by Abu Dhabi at 4.6%, Kuwait at 4.1%, and Bahrain at 3.3%. The Strait of Hormuz accounts for 4.1% of total observations spread across 240 grid cells, reflecting the dispersed nature of transit traffic through the chokepoint compared to the concentrated port activity on the western coast. The Iranian coastline shows limited AIS activity, with Bandar Abbas recording only 1.6% of activity and other Iranian ports such as Bushehr, Jask and Chahbahar at 0.2%, 0.1%, and 0.3% respectively. The disparity in AIS activity between the gulf states and Iran reflects the sheer volume of commercial shipping activity that the Gulf state economies generate compared to their Northern neighbour. The disparity may also reflect differences in AIS compliance between the parties, which will be explored further in section 2.

1.2 Temporal Patterns

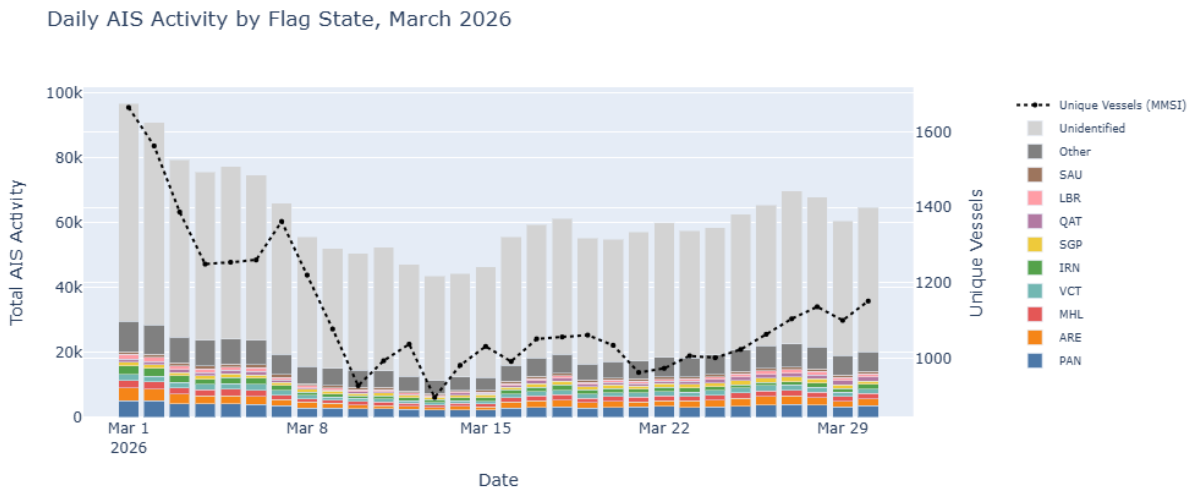


Figure 1.2: Daily AIS Activity by Flag State — March 2026

Figure 1.2 displays the daily AIS activity across the analysed area for March 2026, broken down by flag state. The dotted line represents the daily count of unique vessels transmitting an identifiable AIS signal. The most notable feature is the sharp decline in activity between 1 March and 13 March, where total AIS activity dropped 55% from 96,727 to 43,731 signals and unique identifiable vessels fell 46% from 1,665 to 897. The trough on 13 March follows the major incident on 11 March in which three commercial vessels were struck by projectiles in the Strait (https://www.reuters.com/world/cargo-ship-hit-by-projectile-strait-hormuz-crew-evacuates-2026-03-11/). Activity partially recovered after this trough but did not return to earlier levels, with the post drop period (12–31 March) averaging 57,540 signals per day and 1,036 unique vessels compared to 80,127 signals and 1,392 vessels pre-drop (1-7 March).

Unidentified vessels account for 69.3% of all AIS records across the month. This proportion remained consistent at approximately 69% before and after the drop, indicating that the decline affected identified and unidentified vessels equally rather than one category disproportionately. Of the identified flags, Panama is the most prominent at 5.5% of total activity, followed by the UAE at 3.2%, Marshall Islands at 2.4%, and St Vincent and the Grenadines at 2.2%. Iranian-flagged vessels account for only 2.1% of total AIS hours despite the area covering Iranian territorial waters.

1.3 Vessel Persistence

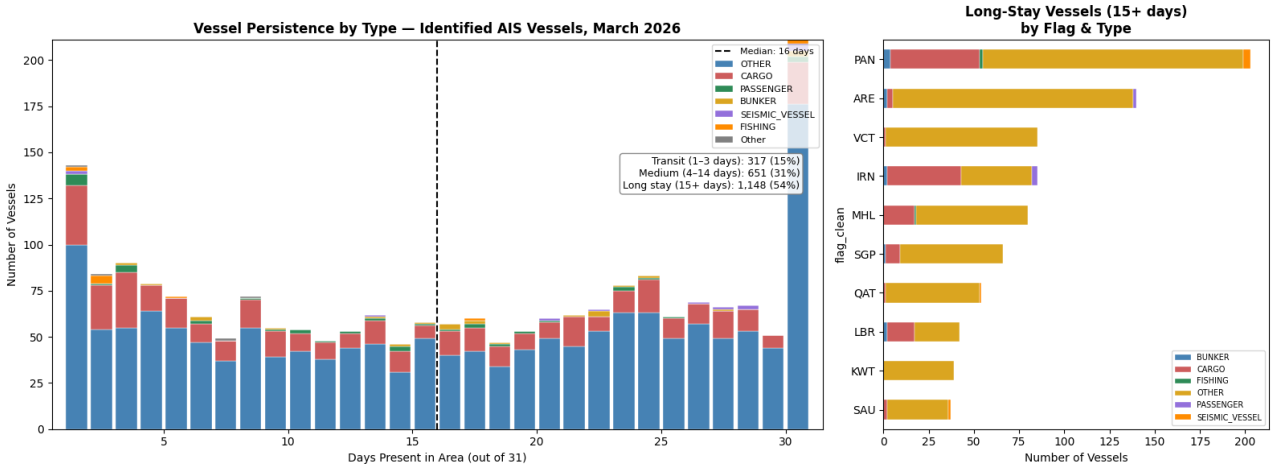


Figure 1.3: Vessel Persistence by Type — Identified AIS Vessels, March 2026

Figure 1.3 shows the persistence of identified AIS vessels in the area during March 2026, measuring the number of days each vessel was present out of 31. Of the 2,116 identified vessels, the median vessel was present for 16 days with a mean of 15.8 days. The distribution shows that 54% of identified vessels (1,148) stayed for a long-stay period of 15 days or more, with a prominent spike at 31 days representing those vessels present for the entire month. Only 15% of vessels managed to leave the area within 3 days. Reporting states that approximately 138 vessels transited the Strait of Hormuz daily prior to war breaking out, but by late March this figure had dropped to as few as 5-6 per day (https://www.abc.net.au/news/2026-04-04/strait-of-hormuz-oil-sailors-stranded-first-supertankers/106525462), suggesting that the low transit numbers in the data reflect those vessels that are unable or unwilling to transit through the area.

The vessel type composition is dominated by the ‘other’ category at 1,616 vessels. GFW classifies vessels into 40 categories, but the AIS presence report only lists 5, with the rest grouped into ‘other’. The next largest category by cargo at 412, with smaller numbers of passenger vessels, bunker vessels, seismic vessels, and fishing vessels at 37, 24, 13, and 10 respectively. The right panel breaks down the long-stay vessels by flag state and type. Panama dominates with over 200 long-stay vessels, the majority of which are classified as bunker and ‘other’ types. The UAE follows with approximately 140 long-stay vessels showing a similar composition. Both Panama and UAE are major ship registration states in the region, so their prominence is expected. Singapore stands out as having a high proportion of its long-stay vessels classified as ‘other’, while Qatar's long-stay vessels are predominantly bunker type, which is consistent with Qatar's role as a major LNG and fuel supplier in the Gulf. Iranian-flagged vessels rank sixth among long-stay flags with a mix of cargo and ‘other’ types, consistent with Iranian commercial and military vessels remaining stationed at home waters throughout the conflict period.

2. Dark Vessel Candidates

Dark vessel candidates are SAR-detected vessels with no corresponding AIS signal at the same grid cell and date. These may represent military vessels, sanctioned vessels with disabled transponders, smuggling operations, or smaller vessels without AIS carriage requirements.

2.1 Spatial Comparison

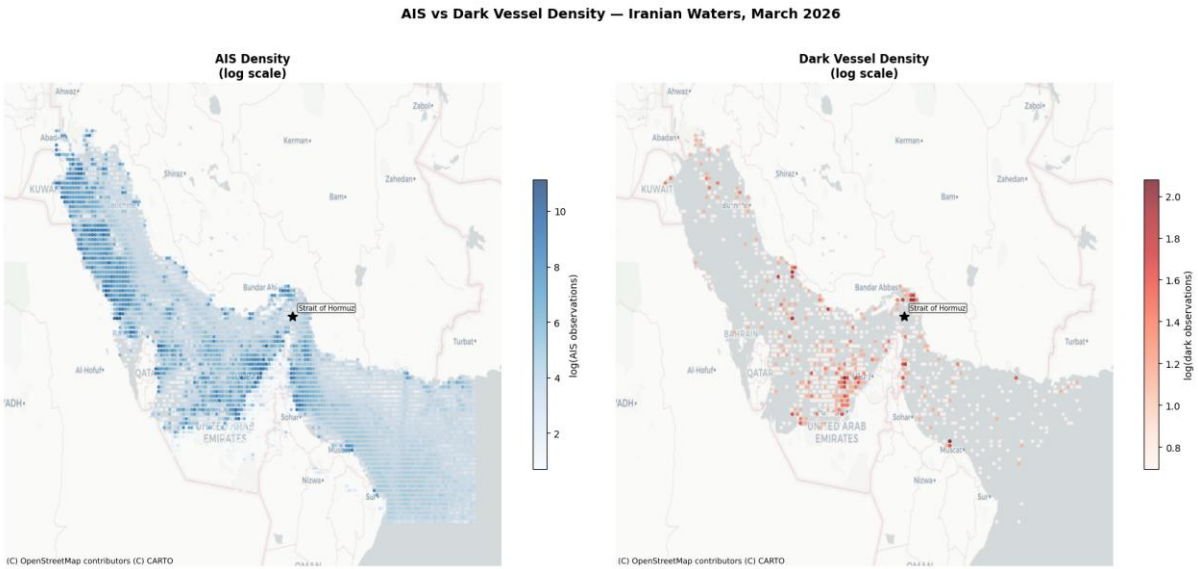


Figure 2.1: AIS vs Dark Vessel Density — March 2026

Figure 2.1 compares the density of AIS vessels against the dark vessel candidates over the analysed area. The left panel reproduces the AIS density seen in Figure 1.1, while the right panel shows the 1,198 dark vessel candidates. The most notable contrast is the geographic shift in activity between the two datasets. AIS activity is concentrated in the west Persian Gulf, which accounts for 66.1% of all AIS observations. Dark vessel activity however is more evenly distributed, with the western Gulf accounting for 48.3% of dark observations and the eastern Gulf 51.7%. This eastward shift in dark vessel activity towards the UAE coastline and Strait of Hormuz is consistent with the operational environment, where vessels in the conflict zone may be disabling their AIS transponders or operating without them.

The Strait of Hormuz and the central Gulf waters stand out as the primary areas where dark vessel activity is most concentrated relative to the surrounding AIS coverage. The densest individual dark vessel cells are located in open water, with the top cells recording 6–7 dark observations per cell. Several of these high-density cells sit in the waters between the Strait of Hormuz and the Omani coast, an area with comparatively sparse AIS coverage. One cell near Muscat at 23.8 degrees North, 58.1 degrees East recorded 7 dark observations against just 2 AIS observations, representing one of only 3 cells across the entire dataset where dark activity exceeded AIS activity. The North-Western Gulf area, despite dominating AIS activity, shows scattered dark vessel detections with no significant clustering, suggesting that the dark vessel population in this area is transient rather than concentrated around specific infrastructure.

2.2 Geographic Concentration

Dark Vessel Candidates by Strategic Zone
Nearest-zone assignment, no double-counting | March 2026

Zone	Candidates	% Total	High Conf (3+)	Max Det.	Days Active
Other / open water	820	68.4%	81	18	28
Strait of Hormuz	124	10.4%	21	17	9
Abu Dhabi (UAE)	56	4.7%	17	7	8
Jebel Ali / Dubai (UAE)	35	2.9%	15	16	4
Fujairah (UAE)	30	2.5%	13	29	8
Bushehr (IRN)	20	1.7%	0	2	6
Doha (QAT)	17	1.4%	1	4	6
Bandar-e Imam Khomeini (IRN)	14	1.2%	0	2	6
Jask (IRN)	13	1.1%	0	2	6
Muscat (OMN)	12	1.0%	0	2	6
Bahrain (BHR)	11	0.9%	2	4	5
Sohar (OMN)	10	0.8%	1	5	4
Abu Musa Island (IRN)	10	0.8%	3	10	4
Kuwait City (KWT)	8	0.7%	1	3	4
Chahbahar (IRN)	7	0.6%	3	4	4
Bandar Abbas (IRN)	6	0.5%	3	5	3
Qeshm / Larak Is. (IRN)	5	0.4%	2	3	4

Figure 2.2: Dark Vessel Candidates by Strategic Zone

Figure 2.2 presents the dark vessel candidates broken down by strategic zone using a nearest-zone assignment to avoid double-counting. 820 dark vessel candidates, accounting for 68.4% of total candidates, fall in open water outside any of the defined zones. Of the predefined zones, the Strait of Hormuz is the clear priority area for dark vessel activity with 124 candidates, 10.4% of total candidates, including 21 high-confidence detections with 3 or more SAR detections. Dark vessel activity was detected in the Strait on 9 days with SAR coverage, with a maximum of 17 detections in a single observation.

The UAE ports collectively account for a significant portion of the remaining candidates, with Abu Dhabi at 56, Dubai/Jebel Ali at 35, and Fujairah at 30. Notably, Fujairah records the highest max number of detections of any zone at 29, and all three UAE zones show a high proportion of high-confidence detections at 17, 15, and 13 respectively. This likely reflects the high volume of vessel activity at these commercial ports combined with the limitations of the 0.1 degree grid resolution, rather than indicating covert operations.

Iranian ports show comparatively few candidates, with Bushehr, Bandar-e Imam Khomeini, and Jask recording candidate counts of 20, 14, and 13 respectively, and none producing high-confidence detections. Bandar Abbas, despite being Iran's primary naval base, records only 6 candidates with 3 meeting the high-confidence threshold. This low figure may be a product of the false negative limitation discussed in the methodology, where dark vessels co-located with AIS-broadcasting vessels in port areas are not flagged. Abu Musa Island is notable as the only Iranian location with a high max detection count of 10, which may warrant further investigation given the island's strategic position in the centre of the Gulf.

3. Satellite Imagery — Bandar Abbas Naval Base

Due to the time constraints of this preliminary investigation, satellite imagery analysis is limited to a single location. Bandar Abbas Naval Base was selected as the target given its strategic significance as Iran's primary naval port and its proximity to the Strait of Hormuz, which was identified in Section 2 as the highest-priority area for dark vessel activity. In a full investigation, this analysis would be expanded to cover additional locations of interest outlined in 2.2. Sentinel-2 optical imagery was acquired for Bandar Abbas via the Copernicus STAC API and cross-referenced with AIS data from the same time window to compare declared vessel presence against what is visually observable.

Figure 3.1: Bandar Abbas Naval Base — Sentinel-2 TCI (10m)
Acquired: 2026-03-18 06:46 UTC | AIS window: 06:00–07:00 UTC

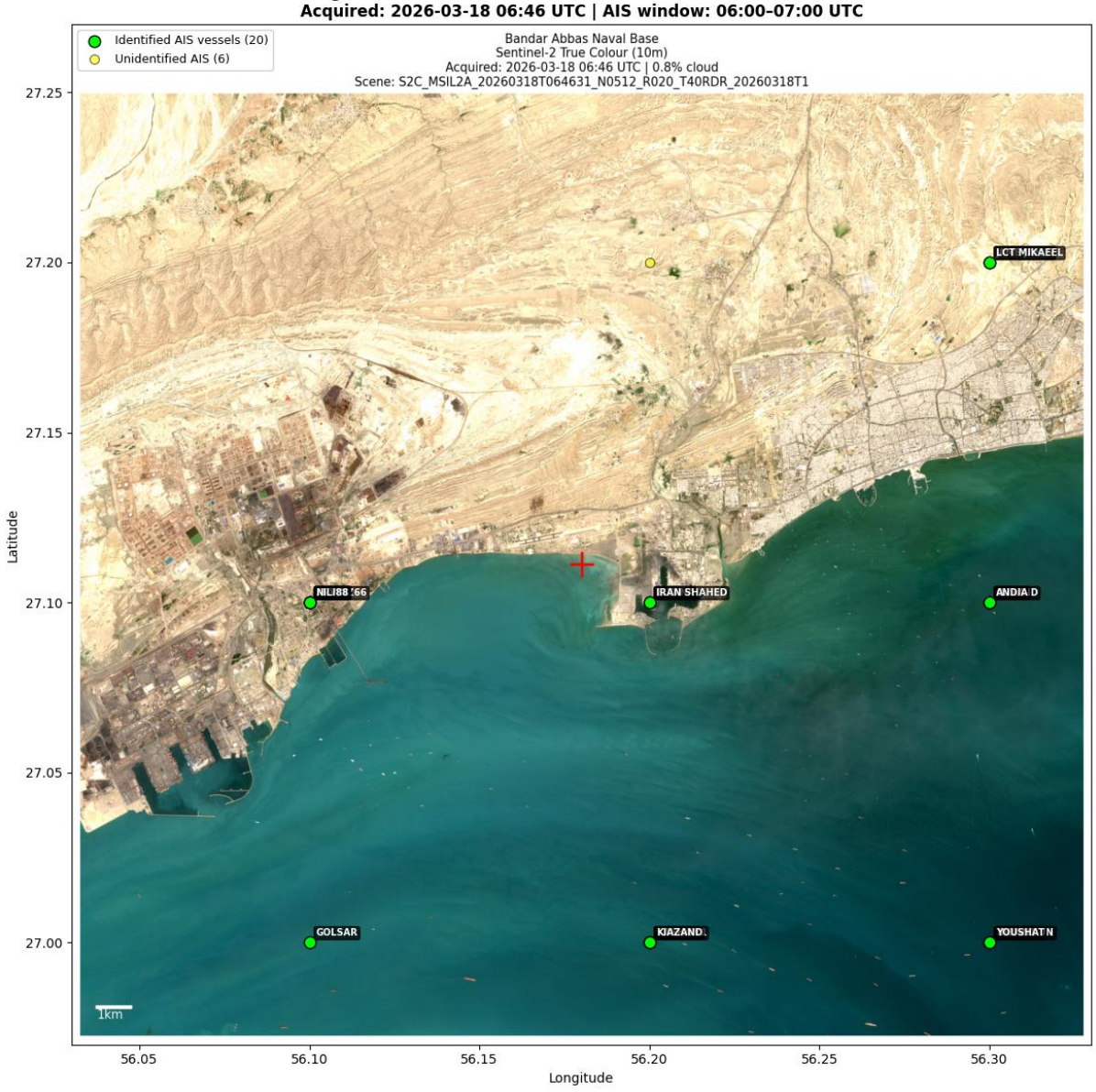


Figure 3.1: Bandar Abbas Naval Base — Sentinel-2 TCI with AIS Overlay, 18 March 2026

Figure 3.1 shows Sentinel-2 True Colour imagery of Bandar Abbas Naval Base acquired on 18 March 2026 at 06:46 UTC with 0.8% cloud cover. This scene was selected as it provided the clearest imagery of the target area during the analysis period, with other available passes having

cloud coverage too high to be reliably analysed. AIS data from the 06:00–07:00 UTC window is overlaid. Of the 82 AIS records in this period, 20 are identifiable vessels and are displayed as green markers. Of these, 19 are Iranian-flagged and one is Panamanian-flagged (NILI88). No dark vessel candidates were recorded in the Bandar Abbas area on this date.

The imagery shows port infrastructure including multiple piers and breakwaters along the coastline, with a large quantity of vessels visible both alongside and in the waters to the south. Visual analysis indicates 100+ vessels just in the observable area. The AIS positions are gridded to 0.1 degrees, approximately 11km, meaning the green markers represent the grid cell a vessel was reported in rather than its precise location. This is evident in the figure where several markers appear over land or clustered at grid intersections rather than aligned with the visible port berths. Despite this limitation, the overlay confirms active port operations at Bandar Abbas during the conflict period, with the identified vessels consistent with activity in Iranian home waters.

The absence of dark vessel candidates at Bandar Abbas on this date is likely a function of the high AIS density in the port area, which as discussed in the methodology masks dark vessels sharing the same grid cell. The overlay of the satellite imagery also highlights a key limitation of the available data at its low resolution. The AIS positions are too coarse to meaningfully match them to individual vessels visible in the 10m imagery. Reliably cross-referencing the data with satellite imagery would require higher resolution geolocation data, such as that available through closed sources.

Findings and Caveats

Key Observations

AIS vessel activity across the Persian Gulf declined sharply during the first two weeks of March, with a 55% drop in total activity and a 46% reduction in unique identifiable vessels between 1 and 13 March, aligning with the escalation of hostilities and the increase in insurance premium costs for shipping. Activity partially recovered but remained approximately 28% below pre-drop levels for the remainder of the month, with 54% of identified vessels present for 15 or more days, indicating a large population effectively unable to transit out of the Gulf.

Dark vessel candidate analysis reveals a geographic distribution that diverges from declared AIS traffic. While AIS activity concentrates in the western Gulf (66.1%), dark vessel activity shifts eastward (51.7% eastern Gulf), with the Strait of Hormuz recording the highest concentration at 124 candidates including 21 high-confidence detections. This is consistent with vessels disabling transponders to avoid targeting in the conflict zone.

Satellite imagery of Bandar Abbas on 18 March confirmed active port operations with over 100 vessels visually observable against only 20 identifiable AIS vessels in the same window. This disparity highlights both the limitations of the low-resolution AIS data and the potential scale of non-AIS-broadcasting activity at Iranian naval facilities.

Caveats and Limitations

The AIS and SAR data are aggregated to 0.1 degree grid cells of about 11km, which directly impacts the dark vessel methodology. Dark vessels sharing a grid cell with AIS-broadcasting vessels in port areas will not be flagged, producing false negatives. SAR coverage is dependent on satellite overpass times and is not continuous, meaning the 1,198 candidates identified represent a lower bound of actual dark vessel presence.

The 69.3% of AIS records classified as unidentified represents a significant data quality limitation. The disparity between the 82 AIS records and the 100+ visually observable vessels at Bandar Abbas further suggests that the GFW dataset may not capture all vessel activity, indicating potential gaps in data collection methodology.

All dark vessel identifications are candidates only, based on open-source data, and would require corroboration from additional sources to be treated as confirmed findings.

Recommended Next Steps

Based on the results of this analytical report and the limitations expressed above, the recommended next steps include: Acquiring higher-resolution AIS data to improve dark vessel identification precision and enable meaningful correlation with satellite imagery. Expand satellite imagery analysis to cover the strategic zones listed in 2.2 but could not be analysed within the time constraints of this report. Correlate temporal patterns in AIS activity with open-source conflict reporting to establish whether specific events correspond to measurable changes in vessel behaviour.